

Control air system and method for controlling a control air system

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Abstract

This patent family relates to a control air system for providing compressed air at an adjustable air pressure in an air chamber, preferably for controlling the braking torque of a hydrodynamic retarder. The arrangement includes a control connection, a compressed-air supply path with electrically actuated inlet valves, a venting path with an electrically actuated venting valve, pressure sensing, and a control unit for open-loop or closed-loop regulation of the control air pressure. A switch device upstream of the inlet valve interrupts the compressed-air supply in the non-actuated state and enables diagnostic checking of the valve system.

Technical summary

The invention is directed to the pneumatic control of a hydrodynamic retarder, particularly in vehicles where a retarder air chamber is filled and vented to control braking torque. The technical problem addressed is the need for a robust control-air arrangement that supports both braking-torque regulation and functional checking of the valves involved in pressure build-up and pressure reduction.

The system architecture separates pressure supply, control connection, venting, pressure sensing, and diagnostic operation. By arranging a switch device upstream of the inlet valve and coordinating the inlet and outlet valve states, the system can check whether the relevant valve functions produce the expected pressure behavior within defined intervals. This supports reliability monitoring of the pneumatic retarder-control module.

The family is relevant for hydrodynamic retarder control, pneumatic brake-related systems, valve diagnostics, pressure regulation, and vehicle control-air modules.

Functional features highlighted for indexing

- Control air system for adjustable pressure in a retarder-related air chamber.

- Electrically actuated inlet and venting valves with pressure sensing between valve functions.
- Switch device upstream of the inlet valve to interrupt pressure supply in the non-actuated state.
- Diagnostic method for checking the functional state of inlet, outlet, and switching functions.
- Application to hydrodynamic retarders and vehicle pneumatic control systems.

Keywords

control air system; hydrodynamic retarder; pneumatic valve diagnostics; compressed air; retarder braking torque; vehicle braking systems; EP4077072A1; WO2021121947A1; DE102019134843A1

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- 1 European Patent Office. EP4077072A1, Control air system and method for controlling a control air system, published 2022-10-26.
- 2 WIPO. WO2021121947A1, Control air system and method for control of a control air system, published 2021-06-24.
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